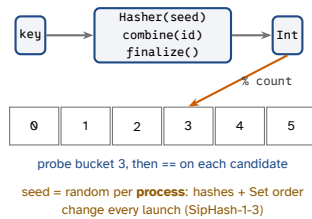


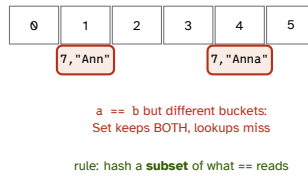
**In one line:** **Equatable** = value equality (`==`); **Hashable** adds `hash(into:)` under one rule —  $a == b \Rightarrow$  **equal hashes** (never the converse); **Comparable** adds a strict total order from **only** `<`. `Set/Dictionary` find a key by *hash first, then ==* — break the contract or mutate a key and elements vanish.

#### A how a lookup works



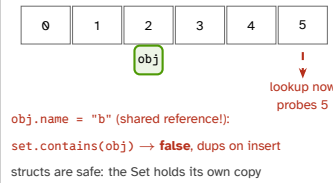
#### B hash uses MORE than ==

`==` compares id; hash combines id + name



#### C mutating a class key in a Set

inserted when `obj.name = "a"` → bucket 2



## How it works

- **Equatable:** reflexive, symmetric, transitive; `!=` is free. **Synthesized** (SE-0185, Swift 4.1) when you declare it and every stored property / associated value conforms; payload-less enums get it without declaring. Never for **classes**. Must be declared in the type's own file.
- **Hashable: Equatable.** Implement `hash(into: inout Hasher)` and combine the same fields `==` reads (SE-0206, Swift 4.2); implementing `hashValue` is deprecated. `Hasher = SipHash-1-3`, randomly seeded per process (HashDoS resistance); `SWIFT_DETERMINISTIC_HASHING=1` fixes the seed for tests. **Never persist** a hash value.
- **Comparable: Equatable.** Only `<` is required; `>`, `<=`, `>=`, `min/max`, `sort()`, ranges `a..b` come free. Must be a strict total order consistent with `==` ( $!(a < b) \ \&\& \ !(b < a) \Rightarrow a == b$ ). Synthesized only for **enums** without raw values whose payloads are `Comparable` — declaration order (SE-0266, 5.3). Tuples (arity  $\leq 6$ ) compare lexicographically: delegate to them.
- **Identifiable** (SE-0261, 5.1): `var id: ID` where ID: `Hashable`; classes get `ObjectIdentifier(self)` by default. **Identity** (which row is this?)  $\neq$  **equality** (did its content change?). SwiftUI keys state, animation and diffing on id.
- **NSObject** bridges: Swift `==` calls `isEqual(_:)`, `hashValue` returns hash. Default `isEqual` is identity. Subclass: override `isEqual(_:)` + `hash` — a custom static `==` or `hash(into:)` is *not* what Set, NSSet or Foundation call.

## Example

```
struct User: Hashable { // identity = id
    let id: Int; var name: String
    static func == (l: User, r: User) -> Bool { l.id == r.id }
    func hash(into h: inout Hasher) { h.combine(id) } // same field
}

struct Version: Comparable {
    let major, minor, patch: Int
    static func < (l: Self, r: Self) -> Bool { // only <
        (l.major, l.minor, l.patch) < (r.major, r.minor, r.patch)
    }
}

final class Node: Hashable { // identity
    static func == (l: Node, r: Node) -> Bool { l === r }
    func hash(into h: inout Hasher) {
        h.combine(ObjectIdentifier(self)) // stable
    }
}
```

Related: value-vs-reference · swift-enums-pattern-matching (synthesis) · strings-and-collections (hash tables) · swiftui-performance-identity · tableview-diffable (item identifiers)

## Which equality for which type?

| Type               | Choose                                                            |
|--------------------|-------------------------------------------------------------------|
| value struct       | synthesize; hand-write only to exclude fields (cache, timestamps) |
| entity (User)      | <code>==/hash</code> on id; Identifiable for SwiftUI              |
| class, identity    | <code>===</code> + <code>ObjectIdentifier</code> (safe to mutate) |
| class, value-based | only if the hashed fields are <code>let</code>                    |
| NSObject subclass  | <code>isEqual(_:)</code> + <code>hash</code> , together           |
| normalised keys    | <code>==</code> and hash on the SAME normal form (lowercased)     |

## Traps

- ! hash combining a field `==` ignores (panel B) — equal values, different buckets.
- ! Mutable class with value-based hash inside a Set/as a dict key (panel C).
- ! `isEqual` overridden, hash forgotten: default hash is identity → broken NSSet/Set.
- ! Set/Dictionary order differs per launch — never assert or persist it.
- ! `Double.nan == .nan` is false: breaks reflexivity, `contains(.nan)` fails, `sort` with NaN is unspecified.
- ! `ForEach(items, id: \.self)` = identity is the whole value: editing a row *changes its identity* (state reset, wrong animation); duplicates collide. Array index as id breaks on insert/reorder.
- ! Hash collisions are legal: unequal values may share a hash.

## Remember

**“Equal  $\Rightarrow$  same hash; hash  $\subseteq$  equality; never mutate a key; `<` buys all of Comparable; id is who, `==` is what.”**